Simulation of High-Efficiency [FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni] Solar Cells Using SCAPS-1D: Optimization and Kinetic Study

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology

in

Chemical Engineering

by

SOHAM KAVATHEKAR **(21BCM0017)**

SUVED MALOKAR **(21BCM0065)**

SOWMYAN NAGARAJ **(21BCM0094)**

Under the Guidance of

Dr. Dharmendra Kumar Bal
(Associate Professor, SCHEME, VIT Vellore)

School of Chemical Engineering

Vellore Institute of Technology, Vellore, Tamil Nadu. 632014

April, 2025

DECLARATION

I hereby declare that the thesis entitled "***Simulation of High-Efficiency [FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni] Solar Cells Using SCAPS-1D: Optimization and Kinetic Study***" submitted by us, for the award of the degree of Bachelor of Technology in Chemical Engineering to Vellore Institute of Technology is a record of bonafide work carried out by us under the supervision of "Dr. Dharmendra Kumar Bal".

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

Date: 30th April 2025

Signature of the Candidate(s) with names

1. Soham Kavathekar

2. Suved Malokar

3. Sowmyan Nagaraj

CERTIFICATE

This is to certify that the thesis entitled "Simulation of High-Efficiency [FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni] Solar Cells Using SCAPS-1D: Optimization and Kinetic Study" submitted by Soham Sudesh Kavathekar (21BCM0017), Suved Sudhir Malokar (21BCM0065), Sowmyan Nagaraj (21BCM0094), School of Chemical Engineering Vellore Institute of Technology, for the award of the degree of Bachelor of Technology in Chemical Engineering, is a record of Bonafide work carried out by him/her under my supervision during the period, 13-12-2024 to 17-04-2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or University. The thesis fulfills the requirements and regulations of the University and, in my opinion, meets the necessary standards for submission.

Place: Vellore

Date: 30th April 2025

Signature of the Guide

Internal Examiner

External Examiner

HoD

Department of Chemical Engineering

Dean

School of Chemical Engineering

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who contributed to the successful completion of this project. First and foremost, I extend my heartfelt thanks to my project guide, Dr. Dharmendra Kumar Bal, for his invaluable guidance, continuous support, and encouragement throughout this project. His expertise and insights were instrumental in helping me navigate complex challenges and inspired me to achieve the best outcomes.

I am also deeply grateful to Prof. Marc Burgelman from the University of Gent for allowing us to use SCAPS-1D, which was an integral component of our research. Its versatility in calculating various performance metrics provided a conclusive result, essential in the completion of this project.

Furthermore, I am sincerely grateful to Dr. Kuldeep Roy, Assistant Professor at NIT Andhra Pradesh, for his unwavering support in our work, and his inputs were instrumental in achieving the results as they stand today.

I would also like to thank the Dean and Head of Department of the School of Chemical Engineering for their support and for providing the resources necessary to carry out this research. Their commitment to fostering a conducive environment for learning and research was crucial to the success of this project.

Lastly, I am grateful to the panel members for their valuable feedback and constructive comments, which greatly enriched the outcomes of this work. Their questions and observations were both insightful and motivating, making a profound impact on my academic journey, and pushing me to explore beyond the scope of this project.

ABSTRACT

The research being conducted used SCAPS-1D simulation software to develop a novel ZnSe/CdTe-based solar cell structure that improves efficiency by employing cost-effective materials. The configuration suggested is FTO/ZnSe/CdTe/ETL/Ni for maximum efficiency. The present investigation aims at enhancing the ZnSe/CdTe combination by selecting the most effective electron transport layer (ETL) from an analysis of materials to enhance device performance. SCAPS- 1D analyses important performance metrics such as circuit voltage (V_{OC}), current density (J_{SC}), fill factor (FF), and efficiency (Ƞ). Thus, multiple simulations have been carried out to figure out the values of efficiency metrics for layer thicknesses. The study also investigates a qualitative comparison of metals for metal back contact function and effect on device performance in terms of efficiency. It was found that Cs_2BiAgI_6 gave a PCE (Power Conversion Efficiency) of 27.99% with Ni as the metal back contact, and this rose to 28.45% when Se was used in Ni's place. The study further included temperature into account while evaluating the performance of this arrangement of solar cells, and it concluded that it displayed a slight reduction in efficiency as temperature rose. This work offers useful insights on determining material characteristics and expediting the development of highly efficient ZnSe/CdTe-based solar cells.

Keywords: Absorption, Efficiency Recombination, SCAPS simulator 1D, Metal back contact, Thickness, Temperature

TABLE OF CONTENTS

	CONTENTS	Page No.
	Acknowledgement	4
	Abstract	5
	Table of contents	6
	List of Figures	7
	List of Tables	8
	Abbreviations	9
	Symbols and Notations	10
1	INTRODUCTION	11
2	LITERATURE REVIEW	13
3	OBJECTIVES AND SCOPE OF WORK	18
4	MATERIALS AND METHODS	15
	4.1 DEVICE CONFIGURATION	19
	4.2 SIMULATION ENVIRONMENT	20
	4.3 PARAMETER OPTIMIZATION STUDIES	20
	4.4 SOFTWARE CALIBRATION AND VALIDATION	21
	4.5 NUMERICAL SIMULATION	21
5	SIMULATION	24
	5.1 SOLAR CELL SIMULATION	25
6	RESULTS AND DISCUSSION	27
	6.1 COMPARATIVE STUDY OF ETLs	27
	6.2 EFFECT OF ETL (CS_2BiAgI_6) THICKNESS	33
	6.3 EFFECT OF CdTe THICKNESS	35
	6.4 EFFECT OF FTO THICKNESS	37
	6.5 EFFECT OF TEMPERATURE ON ETL (Cs_2BiAgI_6) PERFORMANCE	39
	6.6 EFFECT OF METAL BACK CONTACTS ON OPTIMAL CONFIGURATION	41
7	CONCLUSION AND FUTURE SCOPE	43
	REFERENCES	44
	PATENT AND CONFERENCE	49

LIST OF FIGURES

Fig. 1: Novel Arrangement of ZnSe/CdTe-based Solar Cell for Higher Efficiency

Fig. 2: Gantt Chart

Fig. 3.1: Simulation Flow Chart

Fig. 3.2: SCAPS-1D User Interface

Fig. 3.3: SCAPS-1D Problem Interface

Fig. 3.4: SCAPS-1D Layer Interface

Fig 4.1: Effect of thickness on C_{60}

Fig 4.2: Effect of thickness on Cs_2BiAgI_6

Fig 4.3: Effect of thickness on CuO

Fig 4.4: Effect of thickness on Cu_2O

Fig 4.5: Effect of thickness on $PC_{16}BM$

Fig 4.6: Effect of thickness on PCBM

Fig 4.7: Effect of thickness on Sb_2Se_3

Fig 4.8: Effect of thickness on TiO_2

Fig 4.9: Effect of thickness on WS_2

Fig. 4.10: Variations of VOC and JSC for various device thicknesses of Cs_2BiAgI_6

Fig. 4.11: Variations in FF (%) and eta (%) for various device thickness of Cs_2BiAgI_6

Fig. 4.12: Variations in V_{OC} and J_{SC} for various device thicknesses of ZnSe.

Fig. 4.13: Variations in FF (%) and eta (%) for various device thicknesses of ZnSe

Fig. 4.14: Variations in VOC and JSC for various device thicknesses of FTO

Fig. 4.15 Variations in FF (%) and eta (%) for various device thickness of FTO

Fig. 4.16 Variations in V_{OC} and J_{SC} for different temperature ranges

Fig. 4.17: Variations in FF (%) and eta (%) for various device thickness of FTO

Fig. 4.18. Comparison of different metal back contacts with Cs_2BiAgI_6

LIST OF TABLES

Table 1: Literature Review

Table 2: Default physical parameters used to simulate the FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni

Table 3.1: Effect of thickness on C_{60}

Table 3.2: Effect of thickness on Cs_2BiAgI_6

Table 3.3: Effect of thickness on CuO

Table 3.4: Effect of thickness on Cu_2O

Table 3.5: Effect of thickness on $PC_{16}BM$

Table 3.6: Effect of thickness on PCBM

Table 3.7: Effect of thickness on Sb_2Se_3

Table 3.8: Effect of thickness on TiO_2

Table 3.9: Effect of thickness on WS_2

Table 3.10: Qualitative Comparison of ETLs

Table 3.11: Effect of Cs_2BiAgI_6 thickness on solar cell device

Table 3.12: Effect of ZnSe thickness on solar cell device

Table 3.13: Effect of FTO thickness on solar cell device.

Table 3.14: Effect of temperature variance for Cs_2BiAgI_6 at 2μm thickness

LIST OF ABBREVIATIONS

PV – Photovoltaic

TFSC – Thin Film Solar Cell

SCAPS-1D – Solar Cell Capacitance Simulator in 1 Dimension

ETL – Electron Transport Layer

HTL – Hole Transport Layer

FTO – Fluorinated Tin Oxide

V_{OC} – Open Circuit Voltage

J_{SC} – Current Density

FF – Fill Factor

PCE – Power Conversion Efficiency

SYMBOLS AND NOTATION

n - concentration of electrons

p - concentration of holes

J_n - current density of electrons

J_p - current density of holes

G_n - generation rate of electrons

G_p - generation rate of holes

R_n - recombination rate of electrons

R_p - recombination rate of holes

$N_a^-(x)$ - ionized acceptor doping concentration

$N_d^+(x)$ - ionized donor doping concentration

n(x) – free electrons

p(x) – free holes

pt(x) – trapped holes

ε – permittivity

q – electron charge

E – electric field

D – diffusion coefficient

Ψ – electrostatic potential

1. INTRODUCTION

The continuously increasing demand for energy and the adverse environmental impacts of fossil fuel use have intensified the urgent search for alternative energy sources to support sustainable development [1]. Among the various renewable energy sources, solar energy stands out as a dominant and reliable option. Effective harnessing of solar power and its conversion to electricity can address global warming issues and reduce dependency on hydrocarbon-based fuels [2]. Solar energy can be harnessed through solar photovoltaic (PV) systems or solar cells. The primary demands for solar cell technology include the use of abundant, cost-effective, non-toxic materials that offer long operational lifespans and high energy conversion efficiency. At present, natural silicon, which is both widely available and environmentally safe, fulfils over 90% of the global solar photovoltaic market [3]. However, Silicon wafer-based solar cells offer high power conversion efficiency, but their high manufacturing costs, bulkiness, and material consumption limit their impact. These limitations have led to the development of thin-film solar cell (TFSC) technologies, which use alternative materials and fabrication methods. TFSCs have shown promising results, offering high efficiency, cost-effectiveness, environmental sustainability, and the potential for easier industrial-scale production [2]. As the demand for solar energy continues to grow, there is a significant push for the development of alternative materials that are more cost-effective, sustainable, and capable of improving the overall efficiency and scalability of solar technologies [4][5].

One of the biggest challenges faced by the scientific community is to keep up with the rapid rise in demand for energy across the globe. Given the current environmental challenges, it becomes essential to utilize renewable and sustainable sources of energy – particularly, light energy. This is where solar cells, or photovoltaic (PV) cells play a huge role, as they are crucial to achieving net-zero emissions. While solar cell technology has certainly evolved from its primitive first-generation model, which was mainly composed of inorganic, crystalline substances like Silicon (Si) and GaAs (Galium Arsenide), there have been concerted efforts to design newer, second-generation solar cells, utilizing materials that are abundant and can be easily fabricated. Many investigations have proven that Cadmium Telluride (CdTe) is an excellent choice due to its optoelectronic and intrinsic properties. Its cheap cost, relatively harmless nature, ease of fabrication and deposition, and high efficiency strengthen its cause as an absorber layer.

However, CdTe has exhibited poor power conversion efficiency (PCE) on its own. Numerous studies have shown that adding another buffer layer, such as CdS, CdSe, ZnS, or ZnSe, is necessary, as these serve as a focal point to minimize the number of photons lost in the absorber layer. Therefore, this buffer layer needs to possess electrical resistance and minimal surface recombination. While CdS is the most popular metal chalcogenide used in thin-film PV cells, it is harmful and toxic. Thus, ZnSe is promoted as a greener alternative buffer layer to pair with CdTe. The key features of ZnSe include its ability to provide high transmittance at shorter wavelengths and its status as an environmentally friendly buffer layer.

The key objective of this study is to identify and optimize the best electron transport layer that will complement the ZnSe/CdTe configuration. Numerical studies will be carried out using the Solar Cell Capacitance Simulator in One Dimension (SCAPS-1D) Software. This study will investigate the effect of other intrinsic parameters such as the electron transport layer (ETL) material, layer thickness, doping concentration, and metal back contact work function. SCAPS-1D will also be employed to calculate essential data like open circuit voltage (V_{OC}), current density (J_{SC}), fill factor (FF), and efficiency (Ƞ) under standard illumination conditions.

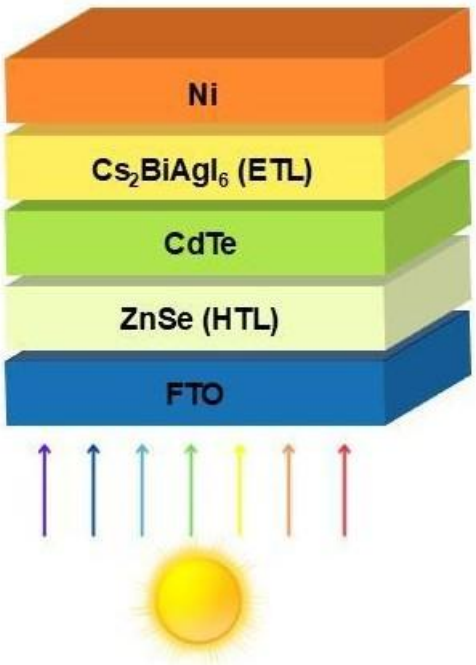


Fig. 1: Novel Arrangement of ZnSe/CdTe-based Solar Cell for Higher Efficiency

2. LITERATURE REVIEW

Table 1: Literature Review

Sl. No	Journal	Author	Title	Year	Outcome
1	European Journal of Engineering Science and Technology	Hasan MallaHasan, Özlem Onay	Investigation of the Effect of Different Factors on the Performance of Several Perovskite Solar Cells: A Simulation Study by SCAPS	2022	Studied the effect of parameters such as ETL and HTL on the performance of solar cells using SCAPS-1D with $MAPbI_3$ showing an efficiency of 24.8%
2	Journal of Physics and Chemistry of Solids	M. El-Mrabet , A. Tarbi , M.A. Hachimi , H. Erguig , T. Chtouki	An optimized design to boost efficiency of CdTe-based solar cell using SCAPS simulator	2024	Studied the optimized combination layer for CdTe-based solar cells using SCAPS-1D.
3	Chemical Physics Impact	A. Bakour, M. Al-Hattab, O. Bajjou, K. Rahmani	Efficiency enhancement of novel ITO/ZnSe/CNTs thin film solar cell	2024	Used SCAPS-1D to calculate the key parameters of a ITO/ZnSe/CNT configuration
4	Journal of Physics and Chemistry of Solids	Khadir A	Boosting the Performance of SnSe-based solar cells through electron and hole transport layers: A qualitative study and perspectives	2024	The study used SnSe as an absorber, with CdS as a buffer layer obtaining an efficiency of 18.65%, and tried other alternatives, using TiO_2 as an ETL obtaining 19.03%

5	Optics and Laser Technology	Hussain Khalid S. M, Faheem Khan, Abdul Ahsan, Khadeeja, Ramachandar amurthy V.K., Abd Razak Bushroa, Subhani Tayyab	Design and Simulation of high efficiency nanostructured TiO_2-Ge solar cell using SCAPS-1D	2024	Simulated TiO2 and Ge multilayered nanostructure coatings on SCAPS-1D, obtaining a power conversion efficiency of 34.1%
6	Journal of Physics and Chemistry of Solids	Çetinkaya, Sameda, Yildirimcan, Saadet	Design and optimization study of all inorganic based double-absorption- layer $CsPbI_3/MoS_2$ heterojunction perovskite solar cells by SCAPS-1D	2024	Calculated Power Conversion Efficiency of an $CsPbI_3/MoS_2$ perovskite solar cell using SCAPS-1D and optimized it from 20.44% to 40.66%
7	Micro and Nanostructures	Selmane, Naceura, Selmane N.; Cheknane, Alia; Hilal, Hikmat S.	A new $CsPbI_2Br$/CuZnSnSSe/Si tandem solar cell with higher than 32 % efficiency	2024	Designed a tandem solar cell using SCAPS-1D featuring $CsPbI_2Br$/CuZnSnSSe/Si, optimizing PCE to 32%

8	Journal of Physics and Chemistry of Solids	Lunge, Omkar Rajendra, Ravidas, Babban Kumar, Bhattarai Sagar, Pandey, Rahul, Madan, Jaya, Roy, Mukesh Kumar, Hossain M Khalid, Samajdar, Dip Prakash	DFT and SCAPS-1D based optimization study of double perovskite $Cs_2AuBiCl_6$ solar cells utilizing different charge transport layers	2024	Optimized $Cs_2AuBiCl_6$ -based perovskite solar cells using various combinations of ETL, with C60 showing 21.16% PCE
9	Journal of Alloys and Compounds	Mishra, Km Pragya; Pandey, Brijesh Kumar; Pandey, Satyabrat	Unveiling the potential of $FASnI_3$ solar cells through advanced charge transport materials: A SCAPS-1D perspective	2024	Optimized $FASnI_3$-based solar cells using copper-based ETLs and HTLs, obtaining a PCE of 32.13%
10	Journal of Physics and Chemistry of Solids	Reza, Md. Selim; Ghosh, Avijit; Wornob, Shafaeit Newaz; Reza, Md. Shamim; Azad, Abdul Kalam;	Tuning the hole transport layer in the Ca_3SbI_3 absorber-based solar cells to improve the power conversion efficiency	2024	Used a cell constructed of SnS_2 and three distinct hole transport layers (HTL) constructed of MoO_3, CBTS, and Cu_2O. Achieved a maximum PCE of 31.65%

		Hossain, Md. Mahfuz; Awwad, Nasser S; Ibrahium, Hala A			
11	Heliyon	Eli Danladi , Philibus M. Gyuk , Nicholas N. Tasie , Anselem C. Egbugha , Debidatta Behera , Ismail Hossain , Ibrahim . Bagudo , Mohammad L. Madugu , Jonathan T. Ikyumbur	Impact of hole transport material on perovskite solar cells with different metal electrode: A SCAPS-1D simulation insight	2023	Identified the best metal contact as Pt due to its best-in-class work function, and obtained a PCE of 25.608% for an HTL device, and 26.229% for a HTL-free device.
12	Crystals	Samer H. Zyoud , Ahed H. Zyoud , Naser M. Ahmed and Atef F. I. Abdelkader	Numerical Modelling Analysis for Carrier Concentration Level Optimization of CdTe Heterojunction Thin Film–Based Solar Cell with	2021	SCAPS-1D Simulations showed that CdTe when paired with a non-toxic metal chalcogenide like ZnSe showed nearly identical efficiencies as traditional HTLs such as CdS.

			Different Non–Toxic Metal Chalcogenide Buffer Layers Replacements: Using SCAPS–1D Software		

3. OBJECTIVES AND SCHEDULE OF WORK

The primary goal of this research is to optimize ZnSe/CdTe-based solar cells by adjusting material combinations and layer configurations using SCAPS-1D. Specifically, the objectives are:

1. To identify the best-performing combination of ETL and HTL layers.
2. To optimize the thickness, Temprature, doping concentration, and defect density of the layers to maximize the PCE.
3. To analyze key performance parameters such as photon conversion efficiency (PCE), open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), and fill factor (FF),

The research project is scheduled over several phases, as shown in the Gantt chart (Figure 3.1). These phases include:

1. Literature review (Month 1st -3rd)
2. Development of simulation methodology (Month 2nd -3rd)
3. Simulation setup and execution (Month 3rd -4th)
4. Analysis and optimization of results (Month 4th -5th)
5. Final report and conclusions (Month 5th – 6th)

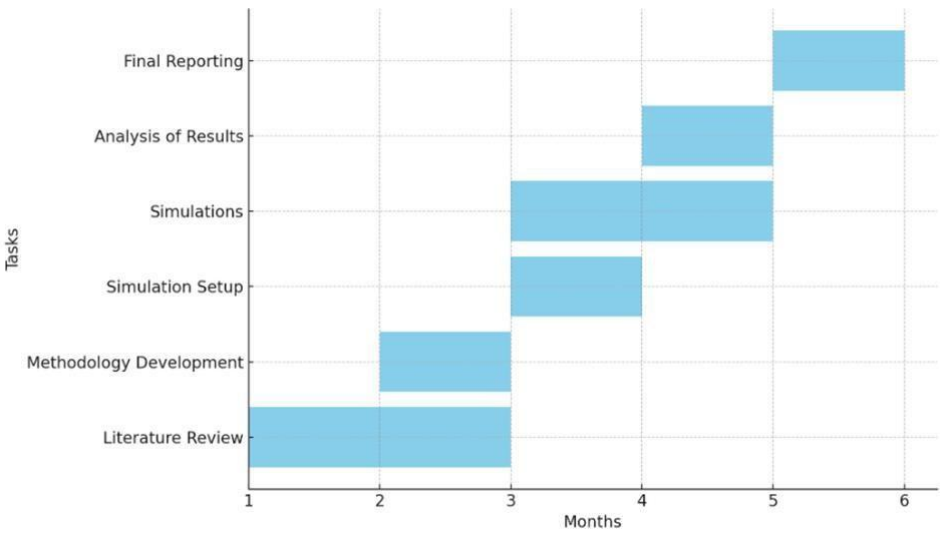


Fig. 2: Gantt Chart

4. MATERIALS AND METHODS

The investigation employed a version of the Solar Cell Capacitance Simulator in One Dimension (SCAPS-1D), a numerical simulation tool designed by Prof. M. Burgelman of the University of Gent, to simulate the performance of a new ZnSe/CdTe-based (PV) device. The literature contends that our primary focus arose from the necessity to augment the device's efficiency and analyse said efficiency concerning material parameters, which were the electron transport layer (ETL) and metal back contact.

4.1 DEVICE CONFIGURATION

The proposed solar cell structure comprises the following layered configuration: **FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni**. Each component of the device plays a critical role in influencing overall performance:

Front Contact (FTO): Fluorine-doped Tin Oxide (FTO) of 0.4 µm thick was chosen as the transparent conductive oxide (TCO) layer because it has an excellent combination of electrical conductivity and optical transparency.

Buffer Layer (ZnSe): A positive hole layer of 0.025 µm-thick Zinc Selenide (ZnSe) was used to avoid using the bottom emissive layer of Cadmium Sulphide (CdS), where ZnSe has a better conductive performance and its transmittance in the visible range is high without having any environmental impact.

Absorber Layer (CdTe): Cadmium Telluride (CdTe), which is p-type and can be 3µm in thickness/size, was the core material that was used for primary photon absorption. CdTe is a direct band gap material with a high absorption coefficient, which makes it appropriate for thin film solar cells.

Electron Transport Layer (ETL): Evaluation of ten different ETL materials of type **Cs_2BiAgI_6**, WS_2, PCBM, WO_3, C_{60}, PC16BM, CuO, Cu_2O, Sb_2Se_3, TiO_2 was carried out. The ETL average thickness is within the range of 0.4 µm and 2 µm. Cs_2BiAgI_6 wins out as the most efficient ETL, giving a power conversion efficiency (PCE) of 27.99%.

Back Contact (Ni): A nickel piece was fixed as the metal back contact since its work function is 5.55 eV. Its selection was based on a balance of high conductivity, thermal stability, corrosion

resistance, and practical availability. Alternative metals, such as Pt, Se, Cu, and Fe, were also studied for comparative performance.

4.2 SIMULATION CONDITIONS

All simulations were conducted under standard test conditions (STC):

Illumination: AM 1.5 solar spectrum at 1000 W/m²

Temperature: 300 K (with temperature variation studies up to 400 K)

Assumptions: Series resistance was assumed negligible, and shunt resistance was taken to be 1000 Ω·cm².

SCAPS-1D solves a set of fundamental semiconductor equations:

Poisson's equation for electrostatic potential distribution

Continuity equations for electron and hole transport

Drift-diffusion equations to model carrier dynamics

These simulations yield key photovoltaic parameters including open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and power conversion efficiency (η).

4.3 PARAMETER OPTIMIZATION STUDIES

To determine optimal configurations, the following variations were systematically explored:

ETL Materials: Comparative analysis of ten ETLs to evaluate their influence on efficiency, Voc, Jsc, and FF. Cs_2BiAgI_6 demonstrated the best overall performance.

Back Contact Metals: Nine metals were assessed, including Se, Pt, Ni, Cu, Fe, Au, Al, W, and C. Selenium yielded the highest theoretical efficiency (28.45%), although Nickel was preferred for its practicality.

Layer Thickness Optimization:

CdTe: Efficiency maximized at 2 µm.

ETL: Efficiency increased with thickness, peaking at 2 µm.

ZnSe: Performance was largely invariant, but 2 µm was adopted for consistency.

FTO: An inverse relationship was observed; thicker FTO led to lower PCE.

Temperature Dependence: Device efficiency was observed to decline beyond 300 K due to increased recombination and decreased carrier mobility.

4.4 SOFTWARE INPUT DATA

The simulation model was calibrated based on literature-reported material properties (band gap, permittivity, doping concentration, mobility, etc.) as detailed in Table 1 of the manuscript. Each simulation scenario was run multiple times to ensure reproducibility and convergence of results.

Table 2: Default physical parameters used to simulate the FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni solar cell configurations [13-19]

Material properties	FTO	ZnSe	p-CdTe	WS_2	PCBM	Cs_2BiAgI_6	WO3	C60	$PC_{16}BM$
d (μm)	0.40	0.025	3.00	0.01	0.01	800	0.05	0.05	0.02
E_g (eV)	3.50	2.9	1.50	1.80	2	1.60	2.6	1.70	2.1
χ (eV)	4.00	4.0	3.90	3.95	3.9	3.90	3.8	3.90	4
ε_r	9.00	10.0	9.40	13.60	3.9	6.50	4.80	4.20	9
N_C (cm^{-3})	2.20×10^{18}	1.50×10^{18}	8.00×10^{17}	1.00×10^{18}	2.50×10^{21}	2.20×10^{18}	2.20×10^{21}	8×10^{19}	2.20×10^{18}
N_V (cm^{-3})	1.80×10^{19}	1.80×10^{19}	1.80×10^{19}	2.40×10^{19}	2.50×10^{21}	1.8×10^{19}	2.20×10^{21}	8×10^{19}	2.2×10^{19}
δ_e (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
δ_h (cm/s)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
μ_e (cm^2/Vs)	20	25	320	100	2×10^{-1}	2×10^0	3×10^1	8×10^{-2}	2×10^1
μ_p (cm^2/Vs)	10	20	40	100	2×10^{-1}	2×10^0	3×10^1	3.5×10^{-3}	1×10^1
N_D (cm^{-3})	1×10^{19}	1.10×10^{18}	0	1.0×10^{18}	2.93×10^{17}	0	6.35×10^{17}	1×10^{17}	1×10^{21}
N_A (cm^{-3})	1.0×10^{14}	0	2.00×10^{14}	0	0	1×10^{15}	0	0	0

4.5 NUMERICAL SIMULATION

SCAPS-1D is a Windows application software created at the University of Gent using National Instruments' Lab Windows/CVI. A solar cell simulation circumstance is represented in an ASCI file, which the SCAPS-1D graphical user interface can read and change entirely. The application is divided into panels, each of which allows the user to specify parameters or display results. SCAPS-1D was chosen for its ease of use, intuitive interface, and advanced features, enabling detailed simulations of solar cell behaviour. The application begins with an 'action panel', where the user may select an operating point (temperature, voltage, frequency, illumination) and a list of calculations to do (C-V, I-V, Q(λ), C-f,). In each computation, the running parameter (λ, f, or V) is adjusted within a given range, while all other parameters

remain fixed at the designated operation point. Solar cell parameters such as current density (J_{SC}), circuit voltage (V_{OC}), efficiency (Ƞ) and fill factor (FF) can be obtained by interpolating the speculated I-V points. Many plots will be displayed using arithmetic scales or linearly scale [8]. The AM1_5G 1 light spectrum, with an intensity of 1000 W/m² (equivalent to one sun or 100 mW/cm²), is used as the light source for solar cell simulations, with an attenuation filter set to 100% transmission unless specified. SCAPS-1D assumes that along the horizontal component, and given the isotropic nature of the material, physical phenomena occurring in this direction have minimal impact on those occurring along the superposition axis. Charge-carrier transport is therefore modelled along the superposition axis by solving Poisson's equation together with the continuity equations for holes and electrons. This simulation tool implements both Poisson's equation and the continuity equations for electrons and holes, as detailed below [9].

$$\frac{d^2}{dx^2}\psi(x) = \frac{q}{\varepsilon_r}\left[p(x) - n(x) + N_D - N_A + N_t\right] \quad [1]$$

$$\frac{1}{q}\frac{d}{dx}J_n = R_n(x) - G_{OP}(x) \quad [2]$$

$$\frac{1}{q}\frac{d}{dx}J_p = G_{OP}(x) - R_n(x) \quad [3]$$

In the equations above, Ψ represents electrostatic potential, q represents electronic charge, ε_r represents relative permittivity, and *n* and *p* indicate free electron and hole concentrations. The densities of the ionised donor and acceptor are denoted by N_D and N_A, respectively, whereas N_t signifies defect density. J_n and J_p signify electron and hole current densities, respectively. R_n and G_{OP} denote the electron-hole pair's recombination and generation rates, respectively [10].

When two separate layers are layered together, a built-in voltage offset (V_{bi}) occurs. The expression for this entity is provided below [10].

$$V_{bi} = E_{g2} - E_{g1} \quad [4]$$

The next layers' band gaps are indicated as E_{g1} and E_{g2}.

To compute the current densities of charge carriers, use the equations given below [10].

$$J_n(x) = q\mu_n E(x) + qD_n\nabla_n(x) \quad [5]$$

$$J_p(x) = q\mu_p E(x) + qD_p\nabla_p(x) \quad [6]$$

The electron and hole mobilities are denoted by μ_n and μ_p, respectively, while the electric field is represented by E. D_n and D_p indicate electron and hole diffusion, respectively.
The equation below yields a plot of current density vs voltage ($J-V$) [10].

$$J = J_{ph} - J_0[e^{(\,{}_gV/\,{}_aK_bT)} - 1] \qquad [7]$$

J_{ph} represents photocurrent density, an is the ideality factor, K_b is the Boltzmann constant, and T is the temperature. J_0 represents the saturation current.

The study specifically focused on planar-junction CdTe-based solar cells arranged in an FTO/ZnSe/CdTe/ETL/Ni structure. To optimize device performance, key parameters were varied, including the metal back contact, wavelength, electron transport layer (ETL), temperature, and layer thickness. By examining a broad range of input values, the optimal conditions for enhancing efficiency were identified. It is largely reliant on three fundamental equations: first, Poisson's equation, following the 1-D continuity equations for electron holes.

In the equations above, Ψ represents electrostatic potential, q represents electronic charge, εr represents relative permittivity, and n and p indicate free electron and hole concentrations. The densities of the ionised donor and acceptor are denoted by N_D and N_A, respectively, whereas N_t signifies defect density. Jn and Jp signify electron and hole current densities, respectively. R_n and G_{OP} denote the electron-hole pair's recombination and generation rates, respectively.

When two separate layers are layered together, a built-in voltage offset (V_{bi}) occurs. The expression for this entity is provided below [10].

5. SIMULATION

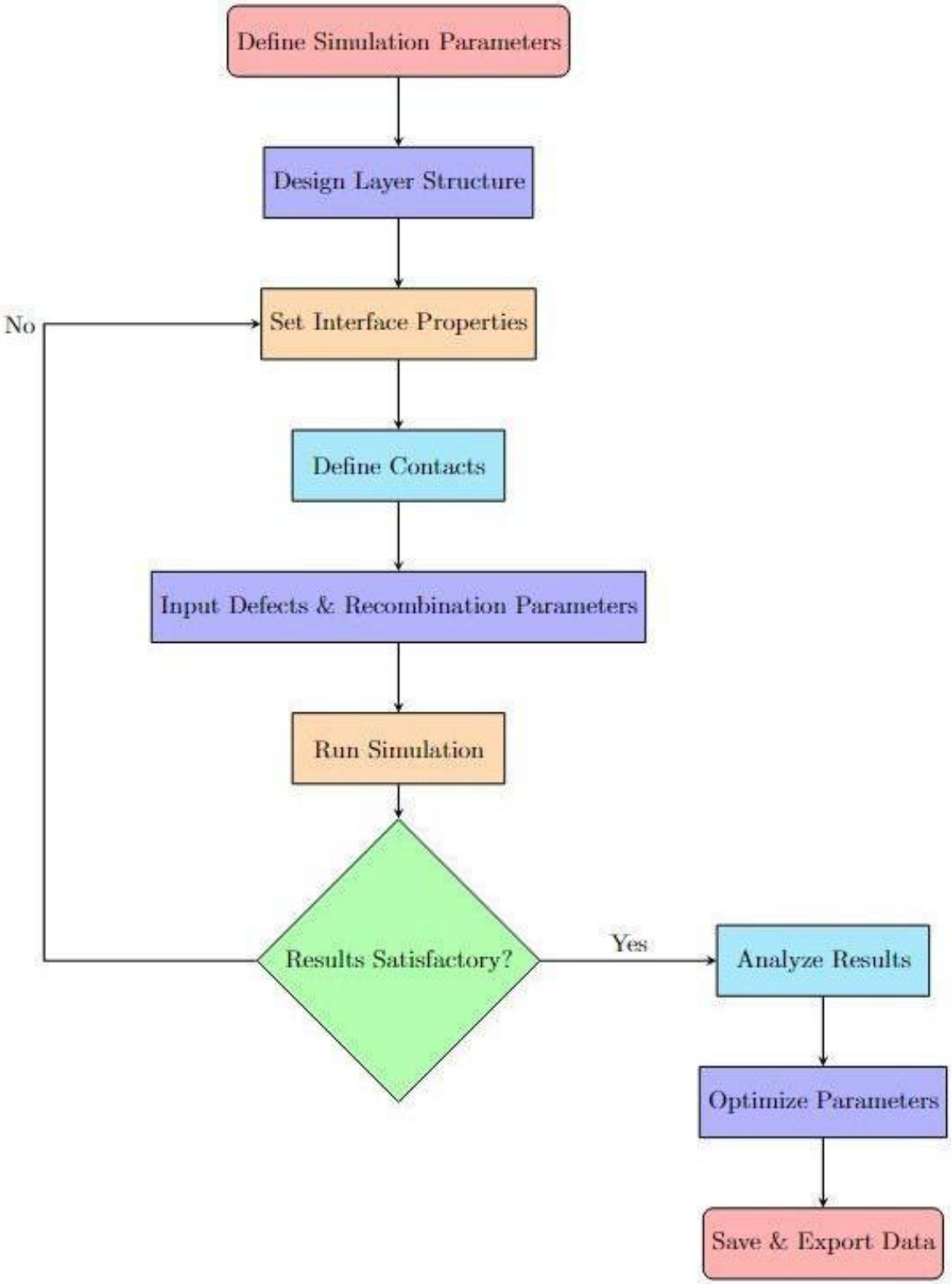


Fig. 3.1: Simulation Flow Chart

5.1 SOLAR CELL SIMULATION

1. Open Software

Fig. 3.2: SCAPS-1D User Interface

2. Problem Interface

Fig. 3.3: SCAPS-1D Problem Interface

3. Calibration Screen

Fig. 3.4: SCAPS-1D Layer Interface

The performance parameters of a solar cell can be calculated by predicting the results of three basic semiconductor equations:

$$dp/dt = \mathrm{Gp} - (p_n - p_{n0})/\tau p - \mathrm{p_n}\mu_\mathrm{p}\, dE/dx + \mathrm{Dp}\, d^2p/dx^2$$

$$dn/dt = \mathrm{Gn} - (n_p - n_{p0})/\tau n - \mathrm{n_p}\mu_\mathrm{p}\, dE/dx + \mathrm{Dn}\, d^2p/dx^2$$

$$d/dx(-\varepsilon(x)d\psi/dx) = \mathrm{q}\,[\,\mathrm{p(x)} - \mathrm{n(x)} + \mathrm{N_d^+}\,(\mathrm{x}) - \mathrm{N_a^-}\,(\mathrm{x}) + \mathrm{pt(x)} - \mathrm{nt(x)}]$$

where the ionized donor doping concentration is represented by $N_a^-(x)$ and the ionized acceptor doping concentration by $N_d^+(x)$.

The free electrons are n(x), the trapped holes are pt(x), and the free holes are p(x). The permittivity is ε, and the electron charge is q.

The electric force field is E, the electron charge is q, the electron diffusion coefficient is D, the rate of generation is G, and the electrostatic potential is ψ.

6. RESULTS AND DISCUSSION

6.1 COMPARATIVE STUDY OF ETLs

ETLs – namely WS_2, PCBM, Cs_2BiAgI_6, C_{60}, WO_3, $PC_{16}BM$, CuO, Cu_2O, Sb_2Se_3, and TiO_2 and their performances were studied employing SCAPS-1D under standard conditions, keeping the rest of the cell configuration constant. The reference configuration is FTO/ZnSe/CdTe/ETL/Ni. From Table 3.10, it can be observed that Cs_2BiAgI_6, C_{60}, and WS_2 have the highest PCEs, yielding 27.99%, 26.59%, and 26.07%, respectively. On the other hand, WO_3 and TiO_2 have the lowest PCEs, yielding 7.37% and 13.99%, respectively. Figs. 4.1-4.9 display the comparison of J_{SC} and V_{OC} for each layer's thickness. With recent developments in the search for sustainable perovskite cells [31], Cs_2BiAgI_6 has emerged as a potential candidate due to its stability, durability, and environmentally benign nature [32,33]. Table 3.10 displays the ETLs used for comparison, with Cs_2BiAgI_6 standing out as the best performer with a record-high PCE of 27.99%, which is reflected by it possessing the highest J_{SC} value compared to the other ETLs.

Thus, the recombination rates are lower as the charge carriers are transported across the cell by the ETL and HTL to their respective junctions. Similarly, C_{60} is also an excellent candidate [34] due to its bandgap energy and high electron mobility, contributing to its high PCE of 26.59%. WS_2 has also garnered some attention from researchers due to its innocuous and ubiquitous nature, yielding 26.07%. PCBM's bandgap alignment is relatively weaker compared to the other three ETLs mentioned above. Similarly, $PC_{16}BM$ has a relatively weaker bandgap, Cu_2O's poor FF implies that there are recombination losses taking place, and weak bandgaps of CuO and Sb_2Se_3 make them mediocre choices. However, TiO_2 and WO_3 displayed very poor fill factors and J_{SC} values, indicating that they possess very poor electron transport properties and suffer from recombination losses.

Figures showed below illustrate the thickness comparison of ETLs with respect to J_{SC}, V_{OC}.

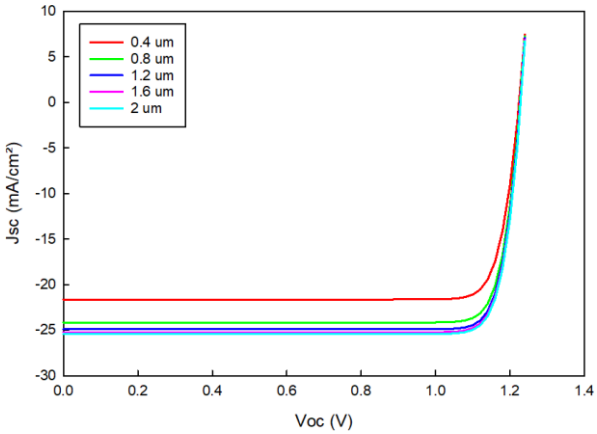


Fig 4.1: Effect of thickness on C_{60}

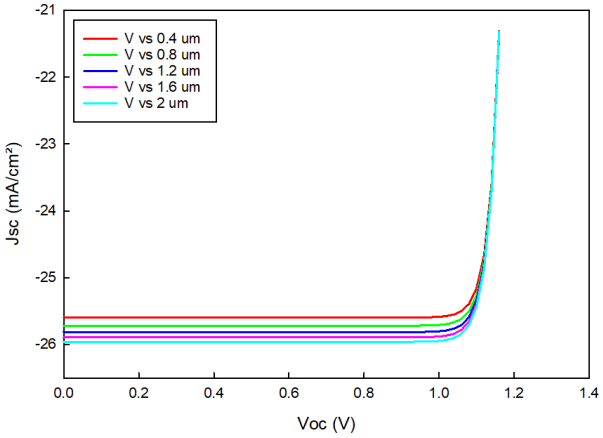


Fig 4.2: Effect of thickness on Cs_2BiAgI_6

Table 3.1: Effect of thickness on C_{60}

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.224625	21.64205287	87.6035	23.2179
0.8	1.227217	24.17250007	87.8529	26.0615
1.2	1.228141	24.92189913	87.9306	26.9135
1.6	1.228671	25.24847288	87.9642	27.2883
2.0	1.229057	25.43454633	87.9798	27.5029

Table 3.2: Effect of thickness on Cs_2BiAgI_6

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.224625	21.64205287	87.6035	23.2179
0.8	1.227217	24.17250007	87.8529	26.0615
1.2	1.228141	24.92189913	87.9306	26.9135
1.6	1.228671	25.24847288	87.9642	27.2883
2.0	1.229057	25.43454633	87.9798	27.5029

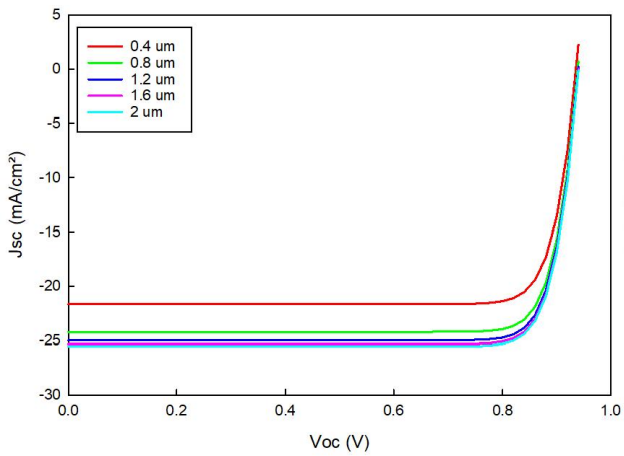


Fig 4.3: Effect of thickness on CuO

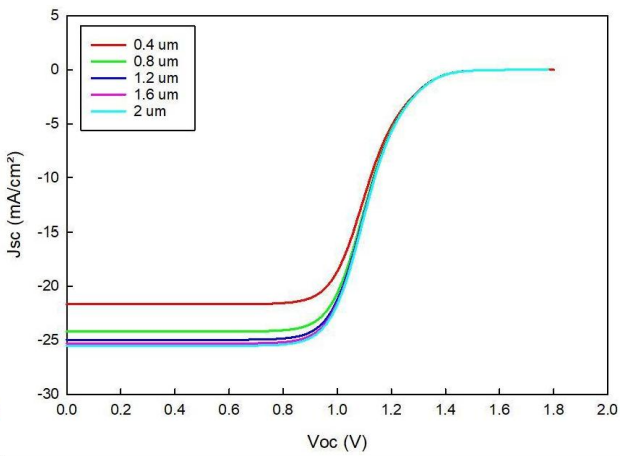


Fig 4.4: Effect of thickness on Cu_2O

Table 3.3: Effect of thickness on CuO

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.224625	21.64205287	87.6035	23.2179
0.8	1.227217	24.17250007	87.8529	26.0615
1.2	1.228141	24.92189913	87.9306	26.9135
1.6	1.228671	25.24847288	87.9642	27.2883
2.0	1.229057	25.43454633	87.9798	27.5029

Table 3.4: Effect of thickness on Cu_2O

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.782022	21.6632	49.9878	19.2974
0.8	1.779271	24.20238	49.9019	21.4891
1.2	1.7755	24.96876	49.9636	22.1499
1.6	1.776468	25.31395	49.9198	22.4486
2.0	1.778945	25.51904	49.843	22.6272

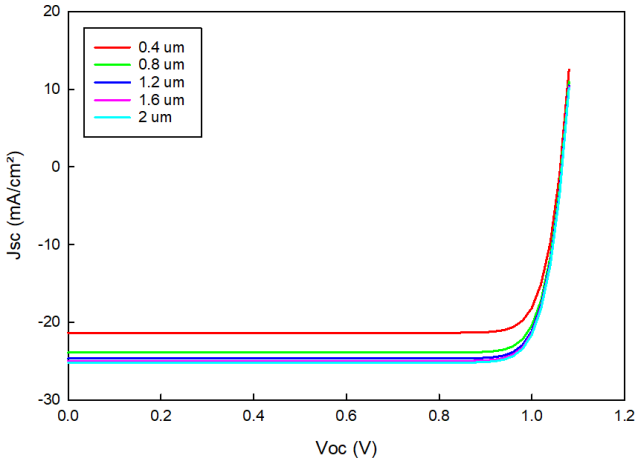


Fig 4.5: Effect of thickness on $PC_{16}BM$

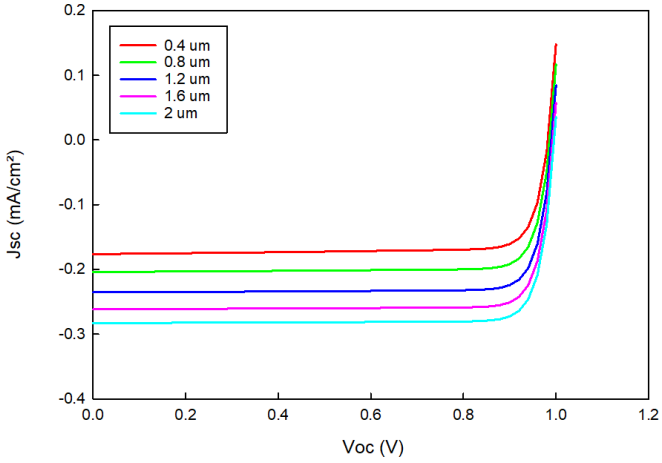


Fig 4.6: Effect of thickness on PCBM

Table 3.5: Effect of thickness on $PC_{16}BM$

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.782022	21.6632	49.9878	19.2974
0.8	1.779271	24.20238	49.9019	21.4891
1.2	1.7755	24.96876	49.9636	22.1499
1.6	1.776468	25.31395	49.9198	22.4486
2.0	1.778945	25.51904	49.843	22.6272

Table 3.6: Effect of thickness on PCBM

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	0.982789	0.17652	83.88	14.55
0.8	0.98713	0.204148	85.8357	17.3
1.2	0.991146	0.235582	86.512	20.2
1.6	0.994157	0.26167	86.8565	22.59
2.0	0.996458	0.282995	87.0622	24.55

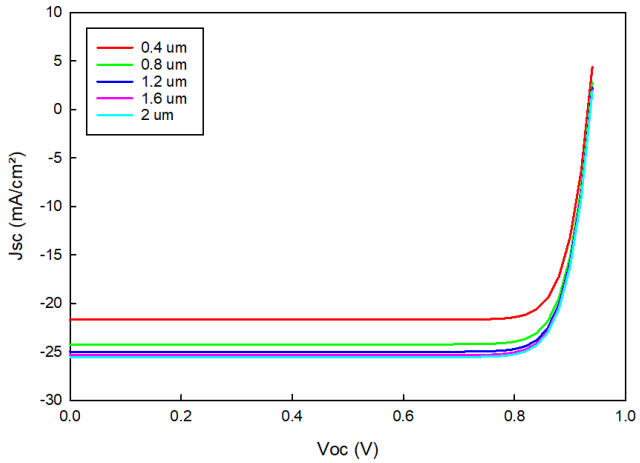


Fig 4.7: Effect of thickness on Sb_2Se_3

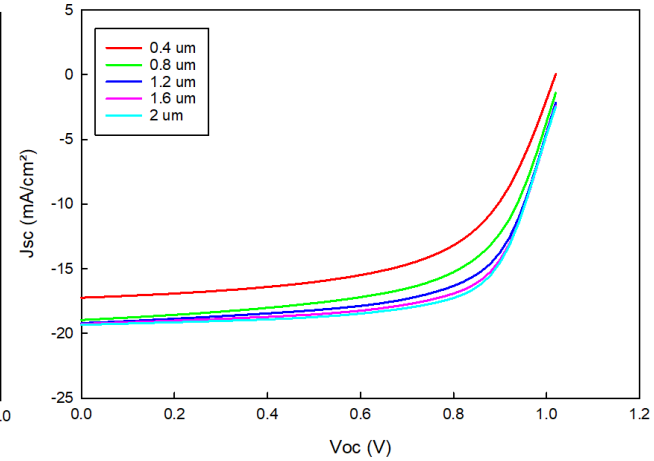


Fig 4.8: Effect of thickness on TiO_2

Table 3.7: Effect of thickness on Sb_2Se_3

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	0.932676	21.67571	85.9714	17.3803
0.8	0.935597	24.22338	85.7612	19.4363
1.2	0.936426	24.99412	85.6933	20.0566
1.6	0.936793	25.34201	85.6616	20.3362
2.0	0.93701	25.54911	85.6425	20.5026

Table 3.8: Effect of thickness on TiO_2

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.019515	17.24049	60.2726	10.5941
0.8	1.033254	18.95896	62.3509	12.2142
1.2	1.041781	19.20799	65.7816	13.1632
1.6	1.046913	19.26773	67.919	13.7004
2.0	1.050487	19.32941	68.904	13.9912

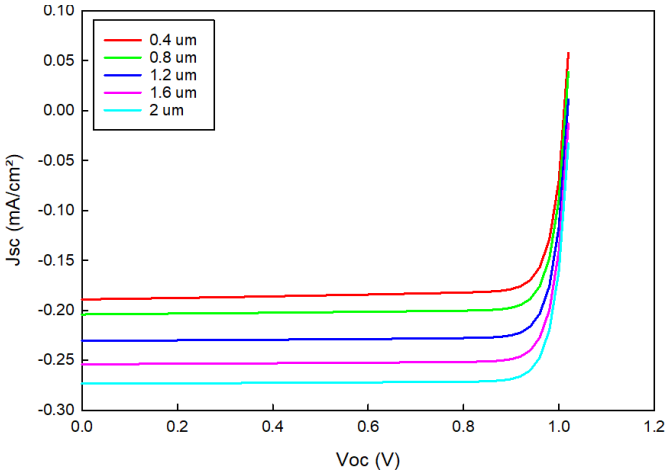


Fig 4.9: Effect of thickness on WS_2

Table 3.9: Effect of thickness on WS_2

Thickness (µm)	**Voc (V)**	**Jsc (mA/cm^2)**	**FF (%)**	**eta (%)**
0.4	1.012204	0.189227	84.5359	0.1619
0.8	1.01494	0.204154	86.4284	0.1791
1.2	1.018623	0.230759	87.0431	0.2046
1.6	1.021263	0.254115	87.3734	0.2267
2.0	1.023101	0.273682	87.5932	0.2453

Table 3.10: Qualitative Comparison of ETLs

ETL	**WS_2**	**PCBM**	**Cs_2BiAgI_6**	**WO_3**	**C_{60}**	**$PC_{16}BM$**	**CuO**	**Cu_2O**	**Sb_2Se_3**	**TiO_2**
V_{OC} (V)	1.1466	1.1174	1.2075	1.3636	1.2278	1.0659	0.9402	1.7789	0.937	1.0505
J_{SC} (mA/cm^2)	25.4335	25.4335	25.962	9.2881	24.637	25.2236	25.5485	25.519	25.5491	19.3294
FF (%)	89.4	89.11	89.28	58.17	87.9	87.09	85.58	49.84	85.64	68.9
eta (%)	26.07	25.33	27.99	7.37	26.59	23.41	20.56	22.63	20.5	13.99

6.2 EFFECT OF ETL (CS_2BiAgI_6) THICKNESS

The effect of ETL thickness on the FTO/ZnSe/CdTe/Cs_2BiAgI_6/Ni cell configuration was investigated using SCAPS-1D. Changing the thickness alters the light-harvesting capacity of the cell, hence making it crucial to identify the optimal thickness. The given configuration was simulated under standard conditions – a solar spectrum AM 1.5 at 300 K and an incident power of 1000 W/m^2, with thicknesses varying from 0.4 μm to 2μm. The investigation also assumed a series resistance [25] at a value close to zero and shunt resistance [26] in the order of 1000 Ω cm^2. Following an increase in thickness as seen in Fig. 4.10, the J_{SC} and V_{OC} of the system increased somewhat linearly finally giving 1.2075 V and 25.9621 mA/cm^2, indicating that a thicker ETL layer absorbs relatively more photons [27], also implying that carriers possess lower recombination rates. Fig. 4.11 portrays the relationship between FF and PCE, where an inverse relationship was observed. While PCE displayed a maximum of 27.99%, the FF linearly dropped. Hence, it is observed that 2 μm was the optimal ETL thickness for the given configuration

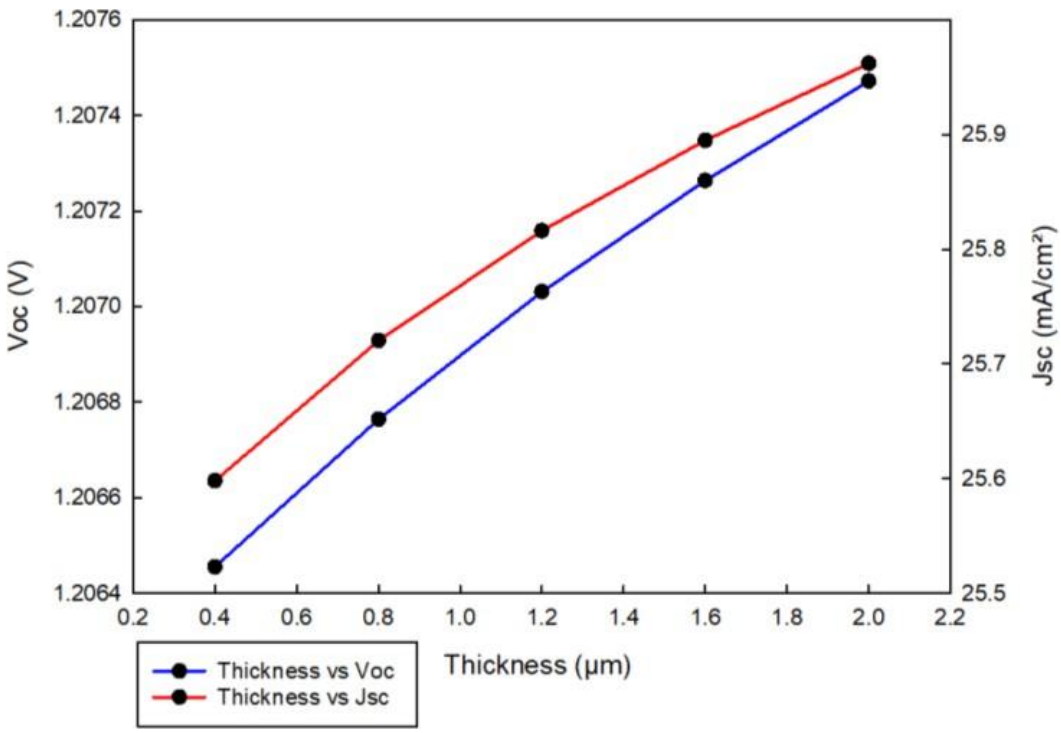


Fig. 4.10: Variations of VOC and JSC for various device thicknesses of Cs_2BiAgI_6

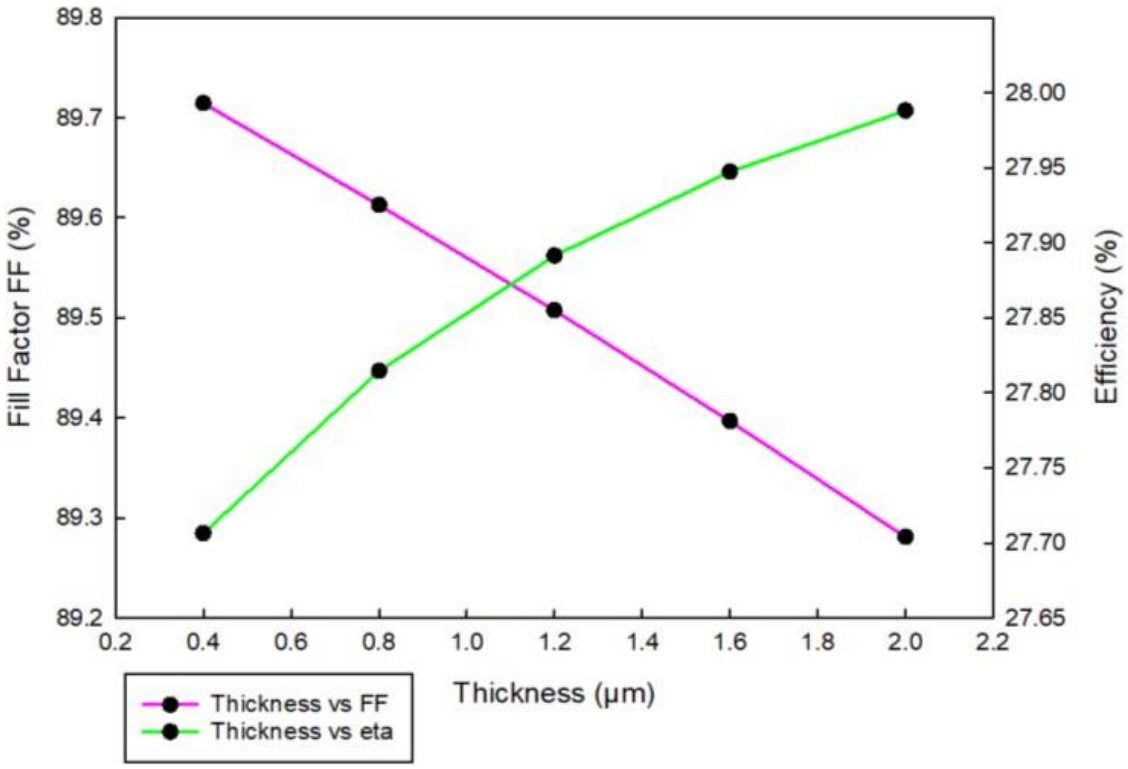


Fig. 4.11: Variations in FF (%) and eta (%) for various device thickness of Cs_2BiAgI_6

Table 3.11: Effect of Cs2BiAgI6 thickness on solar cell device

Thickness (µm)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)	eta (%)
0.4	1.206455	25.59799922	89.7147	27.7064
0.8	1.206764	25.72042826	89.6129	27.8145
1.2	1.207031	25.81620958	89.5075	27.8914
1.6	1.207264	25.89502963	89.3968	27.9474
2	1.207472	25.96207023	89.2809	27.9882

6.3 EFFECT OF CdTe THICKNESS

CdTe's thickness was varied from 0.4 μm to 2 μm to analyze the cell's performance. V_{OC} was observed to rapidly decline to 1.205 V, which can be attributed to the production of a higher series resistance of a thicker CdTe layer [28]. At the same time, J_{SC} steadily grew to 25.7908 mA/cm^2, represented by Fig. 4.12. Keeping in line with the current pattern, Fig. 4.13 reveals that the maximum PCE value was 27.84% obtained at 2 μm. The optimal thickness for CdTe can be identified by the points where the J_{SC} and V_{OC} begin to converge, at 0.8 μm. This thickness maintains optimal photon absorption capacity while maintaining low recombination rates.

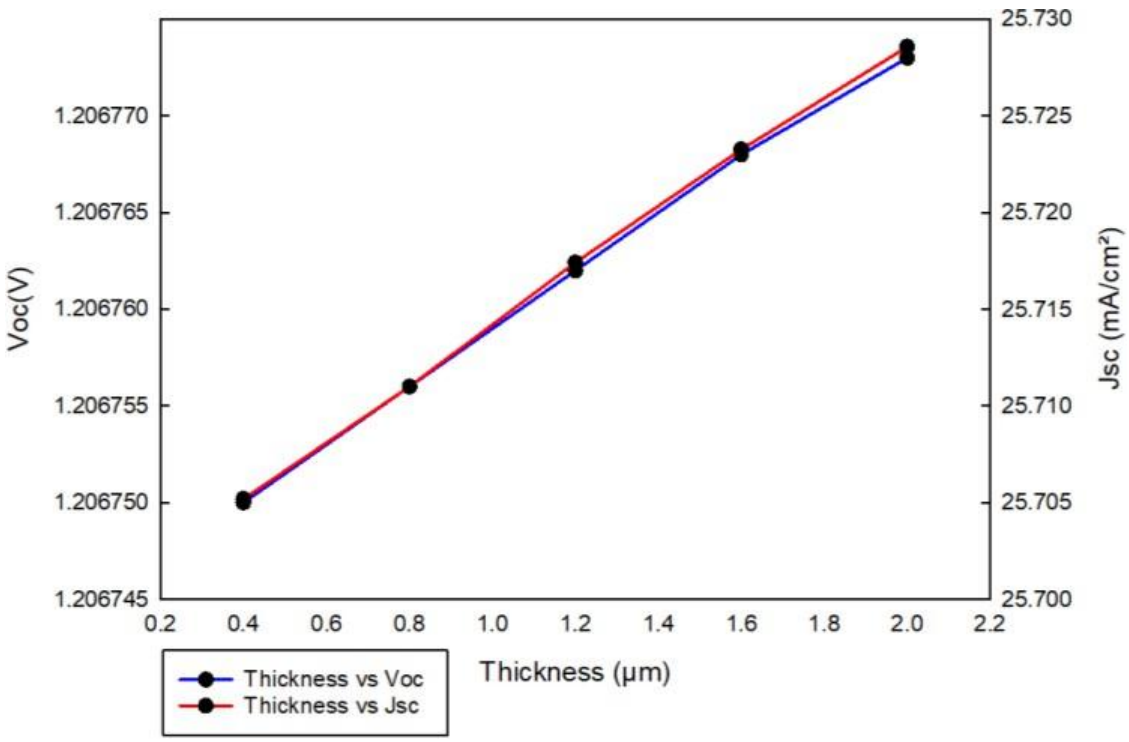


Fig. 4.12: Variations in V_{OC} and J_{SC} for various device thicknesses of ZnSe.

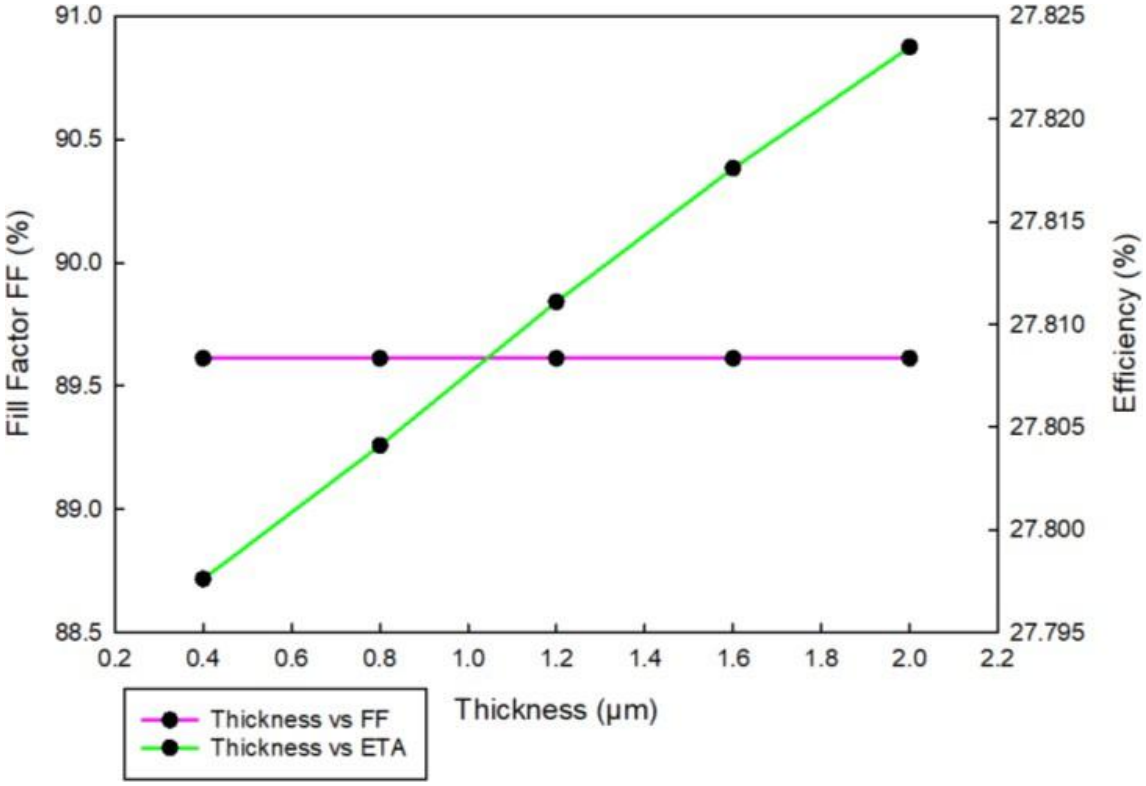


Fig. 4.13: Variations in FF (%) and eta (%) for various device thicknesses of ZnSe.

Table 3.12: Effect of ZnSe thickness on solar cell device

Thickness (µm)	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	eta (%)
0.4	1.20675	25.70518199	89.6128	27.7976
0.8	1.206756	25.71100566	89.6128	27.8041
1.2	1.206762	25.71741716	89.6128	27.8111
1.6	1.206768	25.72326449	89.6128	27.8176
2	1.206773	25.72857298	89.6128	27.8235

.

6.4 EFFECT OF FTO THICKNESS

Upon varying the thickness of FTO from 0.4 μm to 2 μm, it was found that all 4 parameters – V_{OC}, J_{SC}, FF, and PCE all decreased with increasing thickness, which is explained by high recombination rates, thus accounting for the drop in efficiency, witnessing a drop from 27.8858% at 0.4 μm to 26.9556% at 2 μm. Hence, it can be inferred that there is an inverse relationship between PCE and FTO thickness. [27]

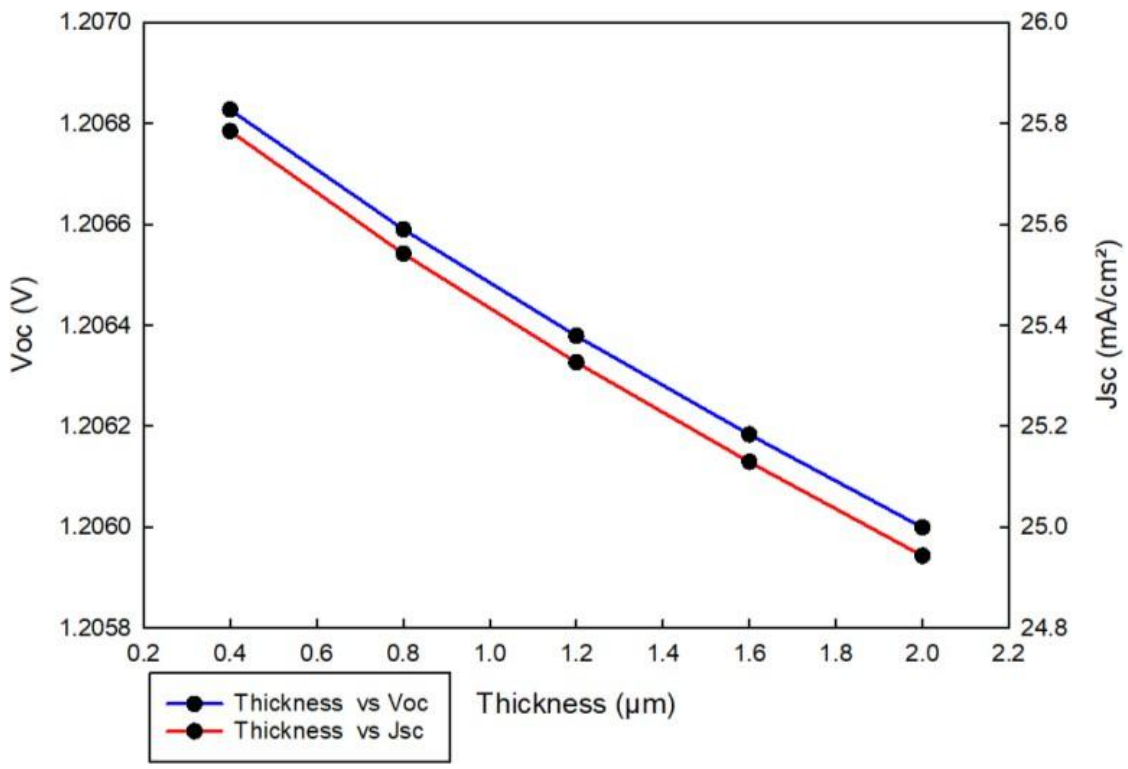


Fig. 4.14: Variations in VOC and JSC for various device thicknesses of FTO

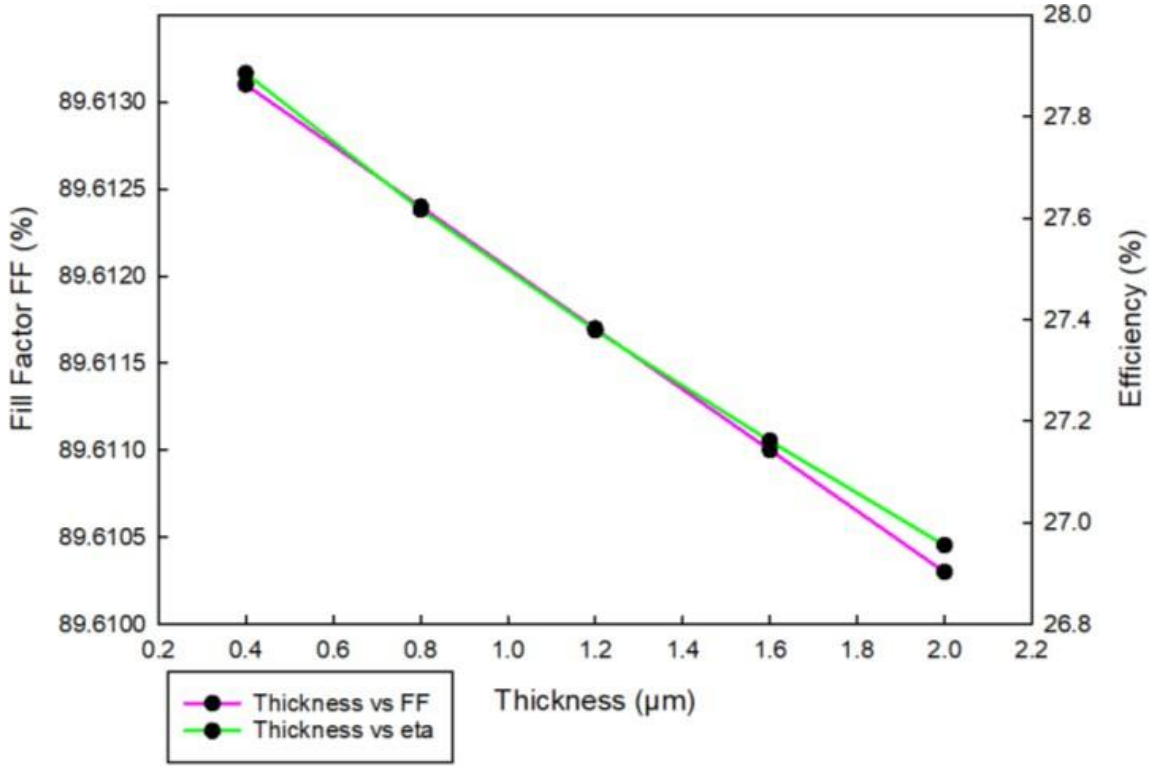


Fig. 4.15 Variations in FF (%) and eta (%) for various device thickness of FTO

Table 3.13: Effect of FTO thickness on solar cell device.

Thickness (µm)	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	eta (%)
0.4	1.206827	25.78494079	89.6131	27.8858
0.8	1.20659	25.54162949	89.6124	27.617
1.2	1.206379	25.32643398	89.6117	27.3793
1.6	1.206183	25.12846122	89.611	27.1607
2	1.205999	24.94270647	89.6103	26.9556

6.5 EFFECT OF TEMPERATURE ON ETL (Cs_2BiAgI_6) PERFORMANCE

A PV cell's performance is heavily influenced by the ambient temperature [29], particularly in industrial applications [30]. In this study, all other simulations were performed at a standard 300 K. However, for this investigation, the temperature varied from 300 to 400 K, and it was observed that V_{OC} gradually declined, whereas J_{SC} plummeted rapidly, before plateauing again, indicating that temperature doesn't affect operations beyond 380 K, as seen in Fig. 4.16. On the other hand, in Fig. 4.17, the PCE and FF steadily dropped with an increase in temperature, indicating that higher temperatures could potentially alter intrinsic properties such as electron and hole mobilities, charge carrier concentrations, and bandgap energies. Excess heat can be the root of carrier collisions, decreasing mobility, and consequently, PCE. [27]

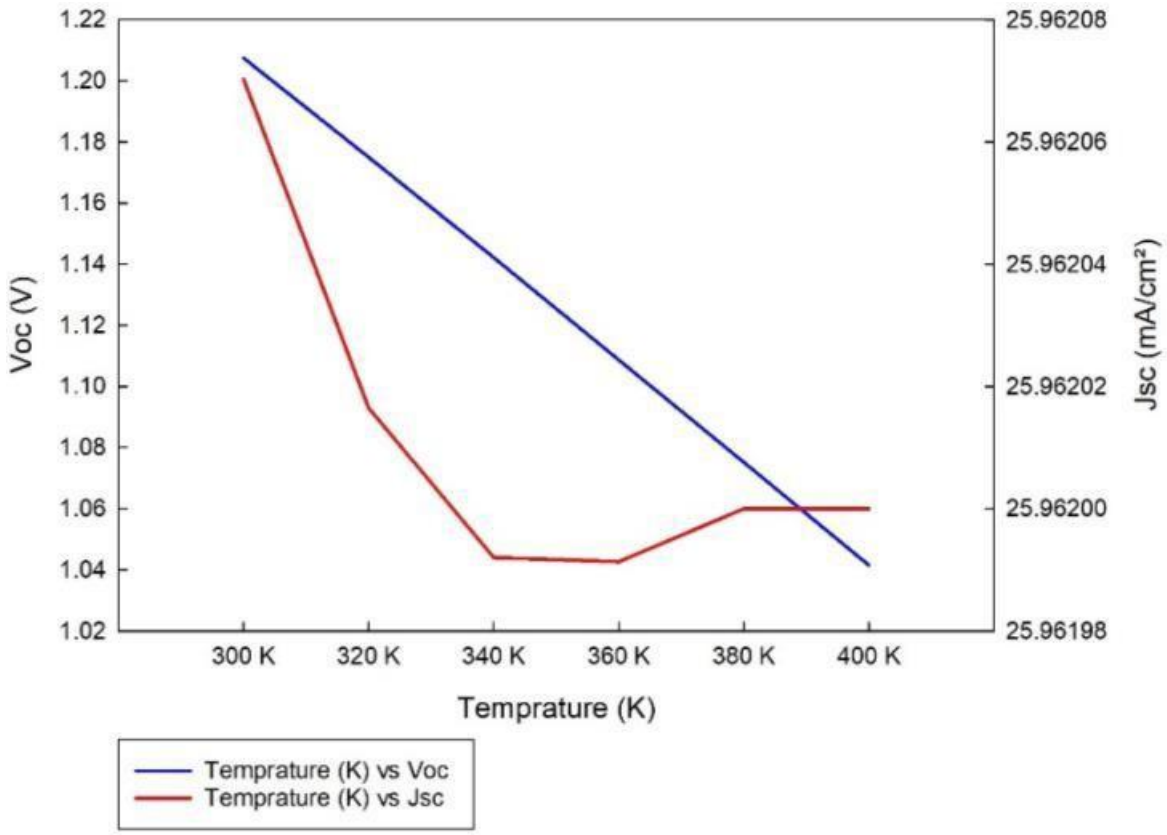


Fig. 4.16 Variations in V_{OC} and J_{SC} for different temperature ranges

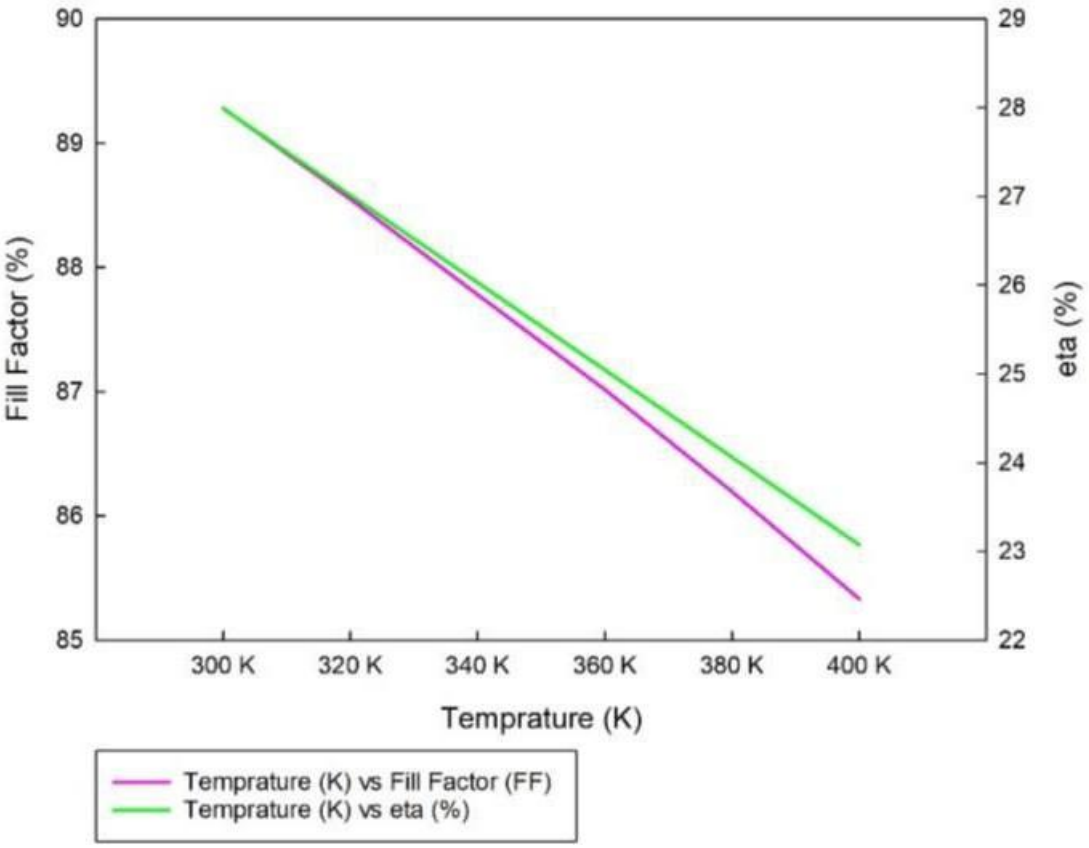
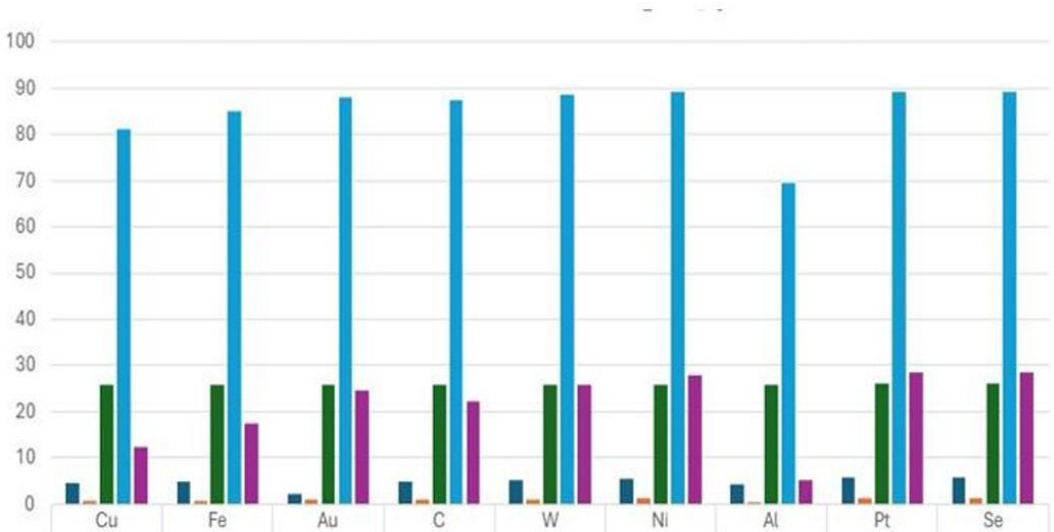


Fig. 4.17: Variations in FF (%) and eta (%) for various device thickness of FTO

Table 3.14: Effect of temperature variance for Cs_2BiAgI_6 at 2μm thickness

Temp. (K)	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	eta (%)
300	1.207472	25.96207023	89.2809	27.9882
320	1.174975	25.96201646	88.5507	27.0121
340	1.142219	25.96199202	87.7825	26.0313
360	1.110000	25.96199134	87.0165	25.0473
380	1.080000	26.00000000	86.2000	24.0608
400	1.040000	26.00000000	85.3000	23.1000

6.6 EFFECT OF METAL BACK CONTACTS ON OPTIMAL CONFIGURATION



	Cu	Fe	Au	C	W	Ni	Al	Pt	Se
Work Function (eV)	4.6	4.8	2.1	5	5.2	5.55	4.3	5.7	5.9
Voc (V)	0.5945	0.7945	1.0876	0.9943	1.128	1.2075	0.2945	1.2296	1.2303
Jsc (mA/cm2)	25.7326	25.7396	25.751	25.7465	25.7677	25.962	25.7231	25.9668	25.9669
FF (%)	81.24	84.86	88.09	87.22	88.48	89.28	69.46	89.06	89.05
eta (%)	12.43	17.35	24.67	22.33	25.72	27.99	5.26	28.44	28.45

Work Function (eV) Voc (V) Jsc (mA/cm2) FF (%) eta (%)

Fig. 4.18. Comparison of different metal back contacts with Cs_2BiAgI_6

This study examines the influence of metal back contacts on the performance of FTO/ZnSe/CdTe/Cs_2BiAgI_6 solar cell configurations. The selected metals, exhibiting p-type semiconductor characteristics when interfaced with n-type semiconductors, were chosen based on their high electron affinity and favourable back-contact interface properties, ensuring optimal charge transport and minimal recombination losses. During the simulations, Ni with a work function of 5.55 eV was used as the metal back contact in all the perovskite solar cell configurations due to its cost-effectiveness, high conductivity, stability, corrosion resistance, thermal durability, and heat dissipation properties [24]. Fig. 2 depicts the effect of the electron work functions of different back-contact metals on the power conversion efficiency of the solar cell.

Metals with higher work functions, such as Pt (5.7 eV) and Se (5.9 eV), demonstrated superior energy band alignment with the p-type semiconductor, leading to reduced recombination losses and improved carrier extraction [21]. This resulted in enhanced efficiency in the simulations.

However, the practical application of these metals is limited due to the rarity of Pt and the toxic nature of Se, making them less viable for large-scale or environmentally sustainable production [22]. As a result, Nickel (Ni) has been identified as the optimal choice for the metal back contact in all simulations. Besides Ni, other metal back contacts, namely Cu (4.6 eV), Fe (4.8 eV), Au (2.1 eV), C (5 eV), W (5.2 eV), Al (4.3 eV), and Pt, were employed and studied with the FTO/ZnSe/CdTe/Cs_2BiAgI_6 configuration.

Nickel, with its moderate work function, provides an excellent balance between efficient charge transport and practical applicability [23]. Its stability at high temperatures, corrosion resistance, and thermal dissipation properties play a crucial role in mitigating heat accumulation at the back contact, which could otherwise lead to performance degradation. This combination of electronic, thermal, and mechanical properties underscores the suitability of Ni as the optimal metal back contact for FTO/ZnSe/CdTe/Cs_2BiAgI_6 solar cells [24]. Thus, Ni offers an ideal compromise, balancing performance with availability and environmental considerations.

7. CONCLUSION AND FUTURE SCOPE

In this study, SCAPS-1D simulation software was employed under standard illumination conditions to analyse and compare the performance of different electron transport layers (ETLs) in photovoltaic (PV) cell design. The versatility and robust features of the SCAPS-1D simulation tool provide a valuable platform for investigating a broad range of material combinations in PV cells. This flexibility enables future investigations to systematically explore and optimize various configurations to achieve enhanced efficiencies.

The investigation highlighted that Cs_2BiAgI_6 achieved the highest power conversion efficiency (PCE) among the various ETLs tested, with an impressive PCE of 27.99% when paired with nickel (Ni) as the metal back contact. This is conclusive to the potential of Cs_2BiAgI_6 as a strong candidate for ETL applications in next-generation solar cells. Notably, when Cs_2BiAgI_6 was used as a reference ETL, selenium (Se) as the metal back contact yielded an even higher efficiency of 28.45%. The study also explored the impact of thickness on the performance metrics of various ETLs. Through these optimizations, the research successfully determined optimal values for key PV parameters, including an open-circuit voltage (V_{OC}) of 1.2075 V, a short-circuit current density (J_{SC}) of 25.962 mA/cm² and a fill factor (FF) of 89.28%. These metrics are instrumental in maximizing the efficiency of PV cells, contributing to the design of high-performing, cost-effective solar technologies.

Furthermore, this research introduced the concept of using a novel alternative buffer layer for PV cell design, specifically highlighting ZnSe as a viable candidate. The choice of ZnSe aligns with both performance improvements and potential cost reductions in the fabrication process, particularly when coupled with an appropriate metal back contact. By incorporating the combination of all these layers, the study paves the way for creating more efficient and economically feasible PV cells that leverage accessible materials without compromising performance. Moreover, it was also found that temperatures beyond ambient conditions had no significant role to play in increasing the efficiency of the device, but rather, it decreased. Excess heat meant that charge carriers collided with each other more often, thus reducing electron and hole mobility, impacting the PCE negatively.

REFERENCES

[1] Rahman, S., Ahmed, S.R. Al, 2021. Photovoltaic performance enhancement in CdTe thin-film heterojunction solar cell with Sb2S3 as hole transport layer. Sol. Energy 230, 605–617. https://doi.org/10.1016/j.solener.2021.10.036

[2] Himanshu, Kamlesh, Suthar, D., Dhaka, M.S., 2023. Numerical simulation of CdSe/ZnTe thin film solar cells by SCAPS-1D: Optimization of absorber layer thickness. Solid State Commun. 371, 115264. https://doi.org/10.1016/j.ssc.2023.115264

[3] Najim, A., Moulaoui, L., Bakour, A., Bajjou, O., Rahmani, K., 2024. High-efficiency numerical modeling of rGO/P3HT solar cells using SCAPS-1D. J. Opt. https://doi.org/10.1007/s12596-024-02112-5

[4] El-Mrabet, M., Tarbi, A., Hachimi, M.A., Erguig, H., Chtouki, T., 2024. An optimized design to boost efficiency of CdTe-based solar cell using SCAPS simulator. J. Phys. Chem. Solids 195, 112287. https://doi.org/10.1016/j.jpcs.2024.112287

[5] Mushtaq, S., Tahir, S., Ashfaq, A., Bonilla, R. S., Haneef, M., Saeed, R., ... & Amin, N. (2023). Performance optimization of lead-free MASnBr3 based perovskite solar cells by SCAPS-1D device simulation. *Solar Energy*, *249*, 401-413. https://doi.org/10.1016/j.solener.2022.11.050

[6] Singh, N., Agarwal, A., & Agarwal, M. (2020). Numerical simulation of highly efficient lead-free all-perovskite tandem solar cell. *Solar Energy*, *208*, 399-410. https://doi.org/10.1016/j.solener.2020.08.003

[7] Basak, A., & Singh, U. P. (2021). Numerical modelling and analysis of earth abundant Sb2S3 and Sb2Se3 based solar cells using SCAPS-1D. *Solar Energy Materials and Solar Cells*, *230*, 111184. https://doi.org/10.1016/j.solmat.2021.111184

[8] Burgelman, M., Nollet, P., & Degrave, S. (2000). Modelling polycrystalline semiconductor solar cells. *Thin solid films*, *361*, 527-532. https://doi.org/10.1016/S0040-6090(99)00825-1

[9]] Mabvuer, F. T., Nya, F. T., Dzifack Kenfack, G. M., & Laref, A. (2023). Lowering cost approach for CIGS-based solar cell through optimizing band gap profile and doping of stacked active Layers─ SCAPS modeling. *ACS omega*, *8*(4), 3917-3928. https://doi.org/10.1021/acsomega.2c06501

[10] Sinha, R. (2024). Design of Aluminum Doped ZnO/Phenyl-C-Butyric Acid Methyl Ester/Tungsten Disulfide (AZO/PCBM/WS2) Heterojunction-Based Device for Applications in Solar Cells and Broadband Self-Powered Photodetectors. *Langmuir*, *40*(13), 7114-7126. http://dx.doi.org/10.1021/acs.langmuir.4c00251

[11] Islam, M.A., Paul, R. A lead-free inorganic Cs2TiX6-based heterostructure perovskite solar cell design and performance evaluation. *Opt Quant Electron* **55**, 957 (2023). https://doi.org/10.1007/s11082-023-05238-1

[12] Burgelman, M., Decock, K., Khelifi, S., Abass, A., 2013. Advanced electrical simulation of thin film solar cells. Thin Solid Films 535, 296–301. https://doi.org/10.1016/j.tsf.2012.10.032

[13] EL-MRABET, M., Tarbi, A., Hachimi, M., Erguig, H., & Chtouki, T. (2023a). *An Optimized Design to Boost Efficiency of CdTe-Based Solar Cell Using Scaps Simulator*. https://doi.org/10.2139/ssrn.4613895

[14] Ahmmed, S., Aktar, A., Rahman, Md. F., Hossain, J., & Ismail, A. B. (2020). A numerical simulation of high efficiency cds/CdTe based solar cell using NIO HTL and zno TCO. *Optik*, *223*, 165625. https://doi.org/10.1016/j.ijleo.2020.165625

[15] Mushtaq, S., Tahir, S., Ashfaq, A., Sebastian Bonilla, R., Haneef, M., Saeed, R., Ahmad, W., & Amin, N. (2023). Performance optimization of lead-free MASNBR3 based perovskite solar cells by SCAPS-1D device simulation. Solar Energy, 249, 401–413. https://doi.org/10.1016/j.solener.2022.11.050

[16] Device Simulation of a Thin-Layer Cssni3based Solar Cell with Enhanced 31.09% Quantum Efficiency. https://doi.org/10.1021/acs.energyfuels.3c00645.s001

[17] Hossain, M. K., Mohammed, M. K., Pandey, R., Arnab, A. A., Rubel, M. H., Hossain, K. M., Ali, M. H., Rahman, Md. F., Bencherif, H., Madan, J., Islam, Md. R., Samajdar, D. P., & Bhattarai, S. (2023). Numerical Analysis in DFT and SCAPS-1D on the influence of different charge transport layers of cspbbr3 perovskite solar cells. Energy & Fuels, 37(8), 6078–6098. https://doi.org/10.1021/acs.energyfuels.3c00035

[18] Mehrabian, M., Taleb-Abbasi, M., & Akhavan, O. (2024). Using CU2O/ZnO as two-dimensional hole/electron transport nanolayers in unleaded FASnI3 perovskite solar cells. Materials, 17(5), 1064. https://doi.org/10.3390/ma17051064

[19] allahyar, soheil, allahyar, sina, & taheri, milad. (2023a). Design Optimization of Efficient CS2BIAGI6-Based Perovskite Solar Cells with Different Charge Transport Layers by Combined DFT, Scaps-1D, and Wxamps Frameworks. https://doi.org/10.2139/ssrn.4341145

[20] Wang, D., Wu, J., Guo, H., Wu, M., Wu, L., Zhang, S., Ao, J., Wang, H., & Zhang, Y. (2020). Tuning the work function of the metal back contact toward efficient Cu2znsnse4 solar cells. Solar RRL, 5(1). https://doi.org/10.1002/solr.202000391

[21] Hall, R. S., Lamb, D., & Irvine, S. J. (2021). Back contacts materials used in thin film CdTe solar cells—a review. Energy Science & Engineering, 9(5), 606–632. https://doi.org/10.1002/ese3.843

[22] Gorai, P., Krasikov, D., Grover, S., Xiong, G., Metzger, W. K., & Stevanović, V. (2023). A search for new back contacts for CdTe solar cells. Science Advances, 9(8). https://doi.org/10.1126/sciadv.ade3761

[23] Aliyu, M. M. (2024). The effect of work function of back contact metals on the performance of CdTe solar cells. Physics Access, 04(01), 51–54. https://doi.org/10.47514/phyaccess.2024.4.1.006

[24] Warepam, D., Dhar, R. S., Singh, K. J., & Biswas, A. (2024). Development and analysis on micro-electrical photovoltaic model of dye sensitized solar cell for optimized performance. Journal of The Institution of Engineers (India): Series D. https://doi.org/10.1007/s40033-024-00689-6

[25] Muduli, S. P., & Kale, P. (2024). Effect of diffusion doping-induced defects on shunt resistance affecting si-nanowire solar cell performance. Journal of Materials Science: Materials in Electronics, 35(6). https://doi.org/10.1007/s10854-024-12190-7

[26] EL-MRABET, M., Tarbi, A., Hachimi, M., Erguig, H., & Chtouki, T. (2023). An Optimized Design to Boost Efficiency of CdTe-Based Solar Cell Using Scaps Simulator. https://doi.org/10.2139/ssrn.4613895

[27] Saikia, D., Bera, J., Betal, A., & Sahu, S. (2022). Performance evaluation of an all inorganic csgei3 based perovskite solar cell by numerical simulation. Optical Materials, 123, 111839. https://doi.org/10.1016/j.optmat.2021.111839

[28] Et-taya, L., Ouslimane, T., & Benami, A. (2020). Numerical Analysis of Earth-abundant CU2ZNSN(SxSe1-X)4 solar cells based on spectroscopic ellipsometry results by using SCAPS-1D. Solar Energy, 201, 827–835. https://doi.org/10.1016/j.solener.2020.03.070

[29] Kanoun, A.-A., Kanoun, M. B., Merad, A. E., & Goumri-Said, S. (2019). Toward development of high-performance perovskite solar cells based on ch3nh3gei3 using computational approach.

Solar Energy, 182, 237–244. https://doi.org/10.1016/j.solener.2019.02.041

[30] Alla, M., Manjunath, V., Choudhary, E., Samtham, M., Sharma, S., Shaikh, P. A., Rouchdi, M., & Fares, B. (2022). Evaluating the potential of lead-free nontoxic Cs2Biagi6-based double perovskite solar cell. Physica Status Solidi (a), 220(2). https://doi.org/10.1002/pssa.202200642

[31] allahyar, soheil, allahyar, sina, & taheri, milad. (2023). Design Optimization of Efficient CS2BIAGI6-Based Perovskite Solar Cells with Different Charge Transport Layers by Combined DFT, Scaps-1D, and Wxamps Frameworks. https://doi.org/10.2139/ssrn.4341145

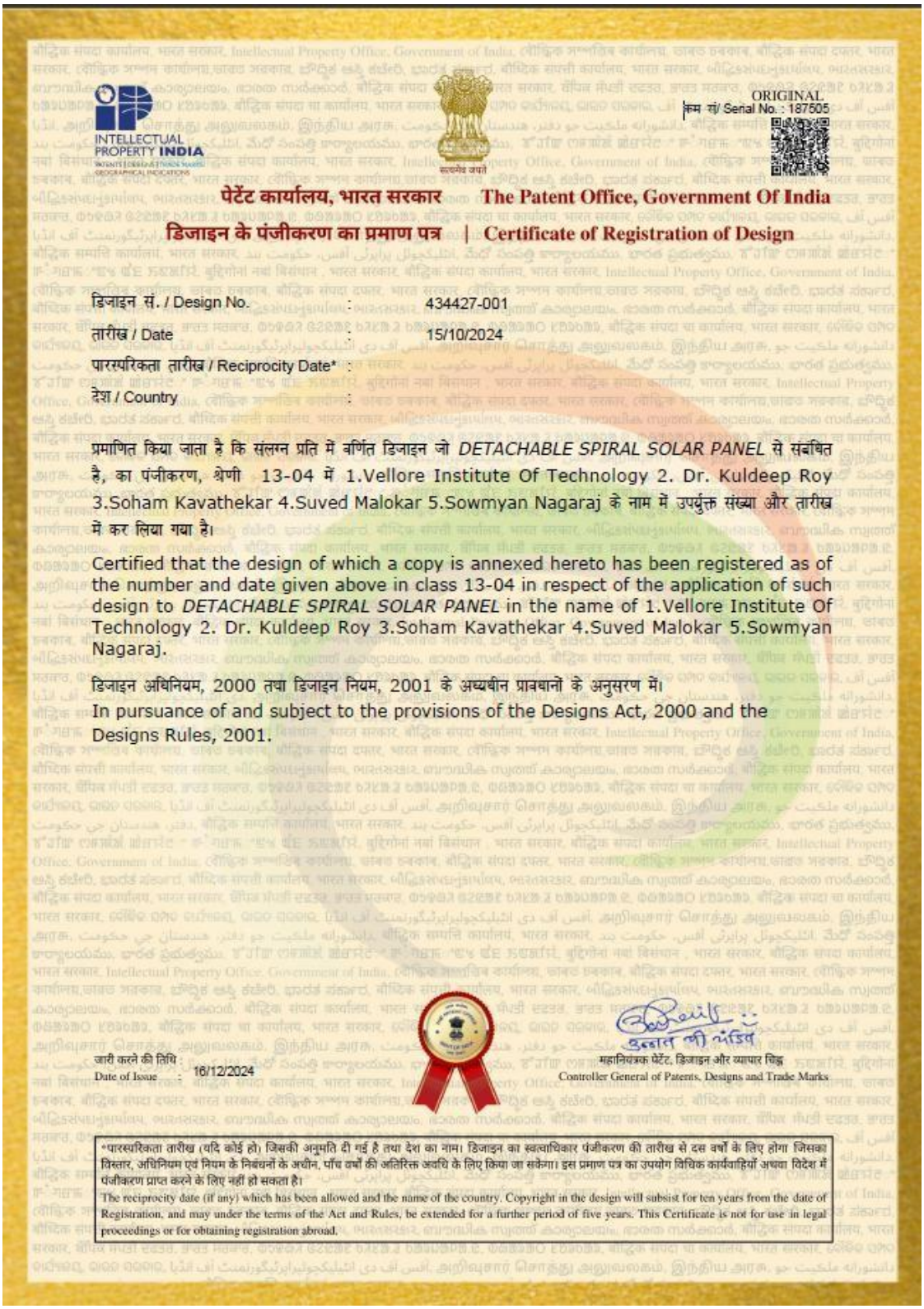
[32] Mishra, A., Khandelwal, B., Singh, B. K., & Shinde, S. (2024). Optimising lead-free CS_2Biagi_6 perovskite solar cells for enhanced performance and stability. Journal of Integrated Science and Technology, 13(2). https://doi.org/10.62110/sciencein.jist.2025.v13.1032

[33] Alathlawi, H. J., Rabhi, S., Hidouri, T., Adawi, H., Makin, F. A., & Alsam, A. A. (2024). Role of AG nanowires: Mxenes in optimizing flexible, semitransparent bifacial inverted perovskite solar cells for building-integrated photovoltaics: A SCAPS-1D modeling approach. Advanced Theory and Simulations. https://doi.org/10.1002/adts.202401004

[34] Jamal Abbas, K., & Bahrami, A. (2024). Investigating the Potential of WO3 and WS2 as CD-Free Buffer Layers in SB2SE3-Based Thin-Film Solar Cells: A Numerical Study with Scaps-1D Software. https://doi.org/10.2139/ssrn.4700811

PATENT AND CONFERENCE

PATENT CERTIFICATE



ORIGINAL

क्रम सं/ Serial No. : 187505

INTELLECTUAL PROPERTY INDIA

पेटेंट कार्यालय, भारत सरकार | The Patent Office, Government Of India

डिजाइन के पंजीकरण का प्रमाण पत्र | Certificate of Registration of Design

डिजाइन सं. / Design No. : 434427-001

तारीख / Date : 15/10/2024

पारस्परिकता तारीख / Reciprocity Date* :

देश / Country :

प्रमाणित किया जाता है कि संलग्न प्रति में वर्णित डिजाइन जो *DETACHABLE SPIRAL SOLAR PANEL* से संबंधित है, का पंजीकरण, श्रेणी 13-04 में 1.Vellore Institute Of Technology 2. Dr. Kuldeep Roy 3.Soham Kavathekar 4.Suved Malokar 5.Sowmyan Nagaraj के नाम में उपर्युक्त संख्या और तारीख में कर लिया गया है।

Certified that the design of which a copy is annexed hereto has been registered as of the number and date given above in class 13-04 in respect of the application of such design to *DETACHABLE SPIRAL SOLAR PANEL* in the name of 1.Vellore Institute Of Technology 2. Dr. Kuldeep Roy 3.Soham Kavathekar 4.Suved Malokar 5.Sowmyan Nagaraj.

डिजाइन अधिनियम, 2000 तथा डिजाइन नियम, 2001 के अध्यधीन प्रावधानों के अनुसरण में।
In pursuance of and subject to the provisions of the Designs Act, 2000 and the Designs Rules, 2001.

जारी करने की तिथि : 16/12/2024
Date of Issue

महानियंत्रक पेटेंट, डिजाइन और व्यापार चिह्न
Controller General of Patents, Designs and Trade Marks

*पारस्परिकता तारीख (यदि कोई हो) जिसकी अनुमति दी गई है तथा देश का नाम। डिजाइन का स्वत्वाधिकार पंजीकरण की तारीख से दस वर्षों के लिए होगा जिसका विस्तार, अधिनियम एवं नियम के निबंधनों के अधीन, पाँच वर्षों की अतिरिक्त अवधि के लिए किया जा सकेगा। इस प्रमाण पत्र का उपयोग विधिक कार्यवाहियों अथवा विदेश में पंजीकरण प्राप्त करने के लिए नहीं हो सकता है।

The reciprocity date (if any) which has been allowed and the name of the country. Copyright in the design will subsist for ten years from the date of Registration, and may under the terms of the Act and Rules, be extended for a further period of five years. This Certificate is not for use in legal proceedings or for obtaining registration abroad.

CONFERENCE

61st ACC 2024 Annual Convention of Chemists and International Conference on Emerging Trends in Chemistry to Revolutionise Indian Chemical Industries for Viksit Bharat @2047 (December 19- 21, 2024)

Title: Best Poster Presentation Award

INDIA'S CENTURY OF CHEMISTRY | JECRC UNIVERSITY | 25 | JMCH

International Conference on Emerging Trends in Chemistry to Revolutionise Indian Chemical Industries for Viksit Bharat @2047

Department of Chemistry, School of Sciences, JECRC University, Jaipur, Rajasthan, (India)

Ref. No. JU/ACC2024/065

CERTIFICATE

This is to certify that

Prof./Dr./Ms./Mr. Soham Kavathekar

of VIT, Vellore presented a paper (oral/poster) on

Optimization and performance - - - - - - SCAPS-1D

Prof. G. D. Yadav, President, Indian Chemical Society | Prof. Victor Gambhir, President, JECRC University | Prof. Sudip Kumar Das, Secretary, Indian Chemical Society | Prof. Sanjay K. Sharma, Convener, JECRC University

61st ACC 2024 Annual Convention of Chemists (ACC 2024) (December 19-21 2024)

INDIA'S CENTURY OF CHEMISTRY | JECRC UNIVERSITY | 25 | JMCH

International Conference on Emerging Trends in Chemistry to Revolutionise Indian Chemical Industries for Viksit Bharat @2047

Department of Chemistry, School of Sciences, JECRC University, Jaipur, Rajasthan, (India)

Ref. No. JU/ACC2024/062

CERTIFICATE

This is to certify that

Prof./Dr./Ms./Mr. Soumyam Nagaraj

of VIT, Vellore presented a paper (oral/poster) on

Optimization and performance - - - - - SCAPS-1D

Prof. G. D. Yadav, President, Indian Chemical Society | Prof. Victor Gambhir, President, JECRC University | Prof. Sudip Kumar Das, Secretary, Indian Chemical Society | Prof. Sanjay K. Sharma, Convener, JECRC University

61st ACC 2024 Annual Convention of Chemists (ACC 2024) (December 19-21 2024)

INDIA'S CENTURY OF CHEMISTRY | JECRC UNIVERSITY | 25 | JMCH

International Conference on Emerging Trends in Chemistry to Revolutionise Indian Chemical Industries for Viksit Bharat @2047

Department of Chemistry, School of Sciences, JECRC University, Jaipur, Rajasthan, (India)

Ref. No. JU/ACC2024/023

CERTIFICATE

This is to certify that

Prof./Dr./Ms./Mr. Suved Malakar

of VIT, Vellore presented a paper (oral/poster) on

Optimization and performance - - - - SCAPS-1D

Prof. G. D. Yadav, President, Indian Chemical Society | Prof. Victor Gambhir, President, JECRC University | Prof. Sudip Kumar Das, Secretary, Indian Chemical Society | Prof. Sanjay K. Sharma, Convener, JECRC University

61st ACC 2024 Annual Convention of Chemists (ACC 2024) (December 19-21 2024)